Thermal Resistance Calculation: Aluminium Plate with Pin-Fin Liquid Cold Plate (Mersen Geometry)

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1 Introduction and Physical Model

The thermal resistance R_{th} [K/W] of a cooled system quantifies the temperature rise from the heat source to the coolant reference per unit of dissipated power:

$$\Delta T = R_{\text{th}}\, P \qquad [\text{K}] \tag{1}$$

The system analysed here consists of:

1. a **5 mm aluminium plate** (dimensions 40 mm × 160 mm) on which the power source (e.g. an IGBT module) is mounted;

2. a **Mersen pin-fin cold plate** attached to the underside of the aluminium plate. The cooling channels are machined as straight fins of depth 10 mm and width 2 mm, separated by 2 mm aluminium walls (pitch 4 mm), running along the 160 mm direction. The coolant is a **30 % ethylene-glycol/water** mixture.

The total thermal resistance is the series of the conduction resistance through the aluminium plate and the cold-plate thermal resistance (which itself incorporates fin conduction, convection, and fin efficiency):

$$\boxed{R_{\text{th,tot}} = R_{\text{cond,Al}} + R_{\text{cold plate}}} \tag{2}$$

2 Geometry

2.1 Aluminium plate

Table 1: Aluminium plate dimensions.

Symbol	Quantity	Value	Unit
L	Length (along fins / flow direction)	160	mm
W	Width (across fins)	40	mm
t	Thickness (heat-flow axis)	5	mm
A	Base area ($L \times W$)	64	cm^2

2.2 Pin-fin cold plate

The channels are rectangular, with the following parameters:

Table 2: Fin/channel geometry of the Mersen cold plate.

Symbol	Quantity	Value	Unit
H_f	Fin (channel) depth	10	mm
w_f	Fin wall thickness	2	mm
w_c	Channel width	2	mm
p	Pitch ($w_f + w_c$)	4	mm
N_f	Number of fins (W/p)	10	–
$\dot{V}$	Volumetric flow rate	24	l/min

Figure 1 shows a schematic cross-section of the cold plate.

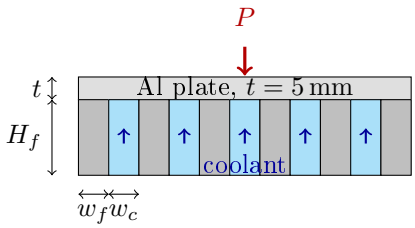


Figure 1: Cross-section of the aluminium plate and Mersen pin-fin cold plate (not to scale, representative subset of fins shown). Grey: aluminium fins and base plate. Blue: coolant channels.

The fin geometry provides a significant wetted-area enhancement over the plain base:

$$A_{\text{fin,total}} = N_f \times 2\,H_f\,L = 10 \times 2 \times 10 \times 10^{-3}\,\text{m} \times 0.160\,\text{m} = 320\,\text{cm}^2 \tag{3}$$

$$A_{\text{base}} = N_f \times w_c \times L = 10 \times 2 \times 10^{-3}\,\text{m} \times 0.160\,\text{m} = 32\,\text{cm}^2 \tag{4}$$

The total wetted area ($A_{\text{fin,total}} + A_{\text{base}}$) is approximately **6.5**× the plain base area, which is the primary reason for the large improvement over an open channel.

3 Conduction Resistance of the Aluminium Plate

$$R_{\text{cond,Al}} = \frac{t}{k_{\text{Al}}\,A} = \frac{5 \times 10^{-3}\,\text{m}}{160\,\text{W/(m}\cdot\text{K)} \times 6.4 \times 10^{-3}\,\text{m}^2} \approx 4.88\,\text{mK/W} \tag{5}$$

This contribution is small but non-negligible at the performance level delivered by the Mersen cold plate.

4 Cold-Plate Thermal Resistance — Manufacturer Data

4.1 Reference data point

Mersen specifies the cold-plate thermal resistance (from the mounting face to the coolant inlet temperature) as:

$$R_{\text{cp}}\big(\dot{V} = 18\,\text{l/min}\big) = 18\,\text{mK/W} \quad \equiv \quad 18\,\text{K/kW} \tag{6}$$

This value already includes fin conduction, convection on all wetted surfaces, and the effect of finite fin efficiency. It is therefore the *recommended* value for system-level thermal design.

4.2 Scaling to a different flow rate

For turbulent flow in pin-fin geometries, the thermal resistance scales approximately as a power law with volumetric flow rate:

$$R_{\text{cp}}(\dot{V}) = R_{\text{cp,ref}} \left(\frac{\dot{V}_{\text{ref}}}{\dot{V}} \right)^{n} \qquad n \approx 0.65\text{–}0.75 \tag{7}$$

Using the midpoint exponent $n = 0.70$ and scaling from 18 l/min to 24 l/min:

$$R_{\text{cp}}(24\,\text{l/min}) = 18\,\text{mK/W} \times \left(\frac{18}{24} \right)^{0.70} \approx 14.7\,\text{mK/W} \tag{8}$$

Table 3 reports the sensitivity to the scaling exponent.

Table 3: Scaled cold-plate resistance at 24 l/min for different exponents.

Exponent n	R_{cp} [mK/W]	Note
0.65	14.93	conservative
0.70	14.72	midpoint (recommended)
0.75	14.51	optimistic

The uncertainty in the scaled value is approximately ±0.2 mK/W, which is negligible in practice.

5 Analytical Fin Model (Cross-Check)

As an independent cross-check, this section derives the cold-plate resistance analytically using the fin efficiency approach.

5.1 Hydraulic diameter and Reynolds number

Each rectangular micro-channel has dimensions $w_c \times H_f = 2\,\text{mm} \times 10\,\text{mm}$:

$$D_h = \frac{4\,w_c\,H_f}{2\,(w_c + H_f)} = \frac{4 \times 2 \times 10^{-3}\,\text{m} \times 10 \times 10^{-3}\,\text{m}}{2\,(2 \times 10^{-3}\,\text{m} + 10 \times 10^{-3}\,\text{m})} \approx 3.33\,\text{mm} \tag{9}$$

The flow is distributed equally among $N_f = 10$ channels. The mean velocity at 18 l/min:

$$\bar{v}_{18} = \frac{\dot{V}}{N_f\,w_c\,H_f} = \frac{3.0 \times 10^{-4}\,\text{m}^3/\text{s}}{10 \times 2 \times 10^{-3}\,\text{m} \times 10 \times 10^{-3}\,\text{m}} \approx 1.50\,\text{m/s} \tag{10}$$

With fluid properties of EG 30 % at 50 °C ($\rho = 1040\,\text{kg/m}^3$, $\mu = 0.82\,\text{mPa}\cdot\text{s}$):

$$\text{Re}_{18} = \frac{\rho\,\bar{v}\,D_h}{\mu} = \frac{1040 \times 1.50 \times 3.33 \times 10^{-3}}{8.2 \times 10^{-4}} \approx 6341 \tag{11}$$

The flow is turbulent ($\text{Re} > 4000$) in both operating conditions.

5.2 Nusselt number (Gnielinski)

$$f = \left(0.790 \ln \text{Re} - 1.64\right)^{-2}, \qquad \text{Nu} = \frac{(f/8)\,(\text{Re} - 1000)\,\text{Pr}}{1 + 12.7\sqrt{f/8}\,(\text{Pr}^{2/3} - 1)} \tag{12}$$

At 18 l/min: $f \approx 0.0353$, $\text{Nu} \approx 51.2$, $h \approx 6918\,\text{W/(m}^2\cdot\text{K)}$.

5.3 Fin efficiency

For a straight rectangular fin of half-thickness $b = w_f/2 = 1\,\text{mm}$ and height $H_f = 10\,\text{mm}$, exposed on both sides (convection coefficient h, aluminium conductivity $k_{\text{Al}} = 160\,\text{W/(m}\cdot\text{K)}$):

$$m = \sqrt{\frac{2h}{k_{\text{Al}}\,w_f}}, \qquad \eta_f = \frac{\tanh(m\,H_f)}{m\,H_f} \tag{13}$$

At 18 l/min: $m \approx 208\,\text{m}^{-1}$, $\eta_f \approx 0.47$.

The fin efficiency is below 50 % because the fins are relatively deep and the convective coefficient is high — tip temperature is noticeably lower than the base.

5.4 Effective area and resistance

$$A_{\text{eff}} = \eta_f\,A_{\text{fin,total}} + A_{\text{base}} \approx 0.47 \times 320\,\text{cm}^2 + 32\,\text{cm}^2 \approx 182\,\text{cm}^2 \tag{14}$$

$$R_{\text{cp,analytical}} = \frac{1}{h\,A_{\text{eff}}} = \frac{1}{6918\,\text{W/(m}^2\text{K)} \times 1.82 \times 10^{-2}\,\text{m}^2} \approx 7.95\,\text{mK/W} \tag{15}$$

Table 4: Analytical fin model results vs manufacturer data.

	18 l/min	24 l/min
h [W/(m^{2}K)]	6 918	9 136
η_f	0.47	0.41
A_{eff} [cm^2]	182	164
$R_{\text{cp,analytical}}$ [mK/W]	7.95	6.69
$R_{\text{cp,Mersen}}$ [mK/W]	18.0	14.7 (scaled)

The analytical model is **optimistic by roughly a factor of 2**, which is expected: it assumes perfect flow uniformity, no entrance effects, ideally straight fins with adiabatic tips, and no contact resistance between the plate and cold plate. The Mersen datasheet value captures all of these real-world losses and should be used for thermal design.

6 Total Thermal Resistance — Summary

Table 5: Total thermal resistance: aluminium plate + Mersen cold plate.

Contribution	18 l/min [mK/W]	24 l/min [mK/W]
Al plate conduction, $R_{\text{cond,Al}}$	4.88	4.88
Mersen cold plate, R_{cp}	18.0	14.7
Total $R_{\text{th,tot}}$	**22.9**	**19.6**

The temperature rise from the mounting surface to the coolant inlet is therefore:

$$\Delta T = R_{\text{th,tot}} \times P \tag{16}$$

- At 18 l/min: $\Delta T \approx 22.9 \times 10^{-3}\,\text{K/W} \times P$ (e.g. 22.9 K for $P = 1\,\text{kW}$)
- At 24 l/min: $\Delta T \approx 19.6 \times 10^{-3}\,\text{K/W} \times P$ (e.g. 19.6 K for $P = 1\,\text{kW}$)

Figure 2 shows the estimated total thermal resistance as a function of flow rate, using the power-law scaling from the Mersen reference point.

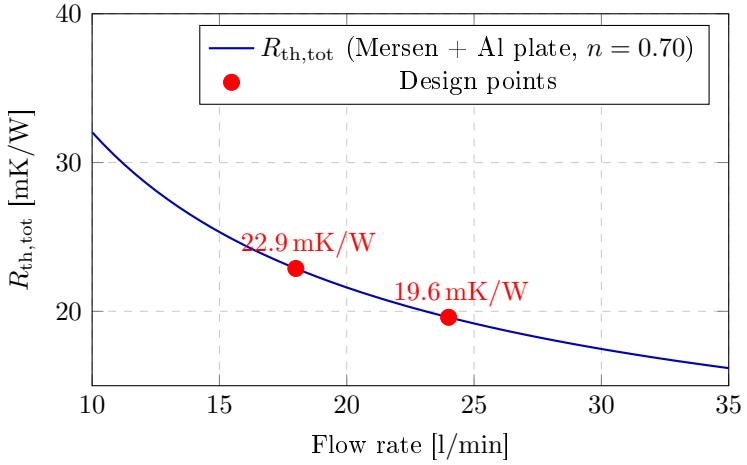


Figure 2: Estimated total thermal resistance vs flow rate. Reference: Mersen at 18 l/min. Power-law scaling with $n = 0.70$.

7 Comparison of Configurations

Table 6 summarises all three configurations analysed in this series of calculations, highlighting the dramatic improvement brought by the Mersen pin-fin cold plate over simple open channels.

Table 6: Comparison of cooling configurations at 24 l/min.

Configuration	h_{eff} [W/(m^{2}K)]	$R_{\text{th,tot}}$ [mK/W]	vs open ch. 5 mm
Open channel, $h_c = 5$ mm	$\approx 8\,300$	23.7	reference
Open channel, $h_c = 12$ mm	$\approx 3\,500$	49.2	$+2.1\times$ worse
Mersen pin-fin (Mersen data)	—	**19.6**	$-17\,\%$ better

The key insight is that the pin-fin structure achieves a better result than the 5 mm open channel despite a larger hydraulic diameter, because the effective heat exchange area is $6.5\times$ larger than the plain base area. The penalty from reduced fin efficiency ($\eta_f \approx 0.47$) is more than compensated by the area enhancement.

8 Assumptions and Limitations

1. **Manufacturer data for cold plate.** The Mersen value of 18 K/kW at 18 l/min is treated as a reliable empirical reference. If the exact product and test conditions differ, the actual value should be verified.

2. **Flow-rate scaling.** The power-law exponent $n \approx 0.70$ is a reasonable estimate for turbulent pin-fin flow; the true value depends on the exact geometry. Where possible, use the manufacturer's flow-rate curve.

3. **Contact resistance.** The thermal interface resistance between the aluminium plate and the cold plate mounting surface is not included. With a good flat contact and thermal grease this is typically $5 - -15 mK/W$ for this area.

4. **Uniform power distribution.** Localised heat sources (e.g. IGBT chips) introduce an additional spreading resistance in the aluminium plate that must be evaluated separately.

5. **Steady-state only.** Dynamic thermal impedance $Z_{\text{th}}(t)$ requires a full RC model including the thermal capacitances of the aluminium plate and the cold plate structure.

9 Conclusions

Using the Mersen manufacturer data as the primary reference, the total thermal resistance of the system (5 mm aluminium plate + pin-fin cold plate) is:

$$R_{\text{th,tot}} \approx 22.9\,\text{mK/W at } 18\,\text{l/min} \qquad R_{\text{th,tot}} \approx 19.6\,\text{mK/W at } 24\,\text{l/min} \tag{17}$$

The pin-fin structure outperforms the open 5 mm channel configuration and is far superior to the open 12 mm channel, confirming that structured surfaces are the correct design choice for high-density power electronics cooling. The remaining dominant resistance is the cold plate itself ($\approx 75\,\%$ of total at 24 l/min); the 5 mm aluminium plate contributes only $\approx 25\,\%$ and is not the bottleneck.